Machine Learning Pipeline for Cardiovascular Disease Classification: Random Forest Optimization with GridSearchCV and Clinical Threshold Recommendation

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***Abstract*—A cardiovascular dataset of 70,000 patients was processed through an end-to-end machine learning pipeline. Five baseline models (Logistic Regression, KNN, Decision Tree, Random Forest, SVM) were evaluated on two split variations (80:20 and 70:30) with 5-fold cross-validation. Random Forest (RF) achieved the highest recall (0.6900) and was selected for tuning using GridSearchCV. The tuning improved AUC from 0.7727 to 0.8057 on the 80:20 split, with an accuracy of 74.06% and F1-Score of 72.16% at the default threshold. At the recommended clinical threshold of 0.40, F1-Score improves to 74.08%, reducing false negatives by 30.5% compared to the default threshold of 0.50. This model is feasible for trial as a screening tool in clinics with limited resources.**

***Index Terms*—Cardiovascular disease, Random Forest, GridSearchCV, clinical threshold, feature selection.**

I. Introduction

Cardiovascular disease (CVD) is the leading cause of death worldwide, with 19.1 million deaths annually according to the WHO [1]. In clinics with limited resources, such as Dr. Siti Rahmawati's clinic, doctors must screen hundreds of patients daily using only blood pressure, cholesterol, and lifestyle data. The main challenge is to identify high-risk patients who need immediate referral, while false negatives (undetected sick patients) can be fatal.

Machine learning (ML) has proven effective for early CVD prediction [2]. In particular, ensemble algorithms like Random Forest (RF) excel in handling heterogeneous features and avoiding overfitting through tree aggregation [3]. Besides model selection, hyperparameter optimization and decision threshold calibration are crucial for clinical implementation. Without proper threshold calibration, even a good model can produce too many harmful false negatives.

The contributions of this paper are: (1) a feature engineering pipeline with medical validity cleaning and correlation-based selection; (2) evaluation of five baseline models with two split ratios; (3) RF tuning using GridSearchCV; (4) threshold analysis to provide clinical recommendations that minimize false negatives.

II. Methodology

A. Dataset

The Cardiovascular Disease Dataset from Kaggle [5] contains 70,000 samples with 11 features: age, gender, height, weight, systolic blood pressure (ap_hi) and diastolic blood pressure (ap_lo), cholesterol, glucose, smoking, alcohol, physical activity. All features are numerical, so no additional encoding is needed. The target is binary (1 = CVD, 0 = healthy). The target distribution is fairly balanced (50.03% healthy, 49.97% CVD). No missing values were found. Thus, the data is ready for preprocessing.

B. Preprocessing and Feature Engineering

Medical cleaning was performed by removing data with physiologically implausible values (ap_hi 60–250 mmHg, ap_lo 40–200 mmHg, height 100–220 cm, weight 30–180 kg). A total of 1,261 rows (1.8%) were removed, leaving 68,739 samples with balanced classes (50.51% healthy, 49.49% CVD). Patient ages ranged from 29.6 to 64.9 years (mean 53.3 years), and mean systolic pressure was 126.6 mmHg.

Feature creation added five new features: BMI, pulse pressure (ap_hi − ap_lo), mean arterial pressure (MAP = ap_lo + pulse pressure/3), age group (4 categories), and BMI category (4 categories). These features are clinically known to be related to CVD risk.

Feature selection used Pearson correlation on the 17 features (Figure 1) to identify pairs with $|r| > 0.85$. Redundant members were removed: MAP (r=0.93 with ap_lo), age group (r=0.91 with age), BMI category (r=0.85 with BMI), and weight (r=0.85 with BMI). Low-correlation but clinically relevant features (gender, smoking, alcohol, height) were retained. A 5-fold logistic regression comparison showed stable performance: AUC 0.7904 (12 features) vs. 0.7903 (16 features), F1 0.7078 vs. 0.7074.

Standardization used a StandardScaler fitted only on the 80:20 training split to prevent data leakage, resulting in 12 final features.

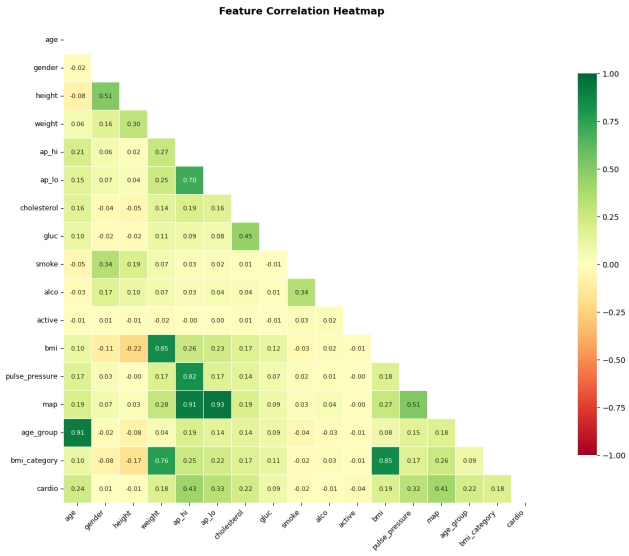

Fig. 1. Pearson correlation heatmap. Dark cells indicate high correlation (>0.85) guiding redundant feature removal.

C. Baseline Models

Five baseline models were trained: Logistic Regression (LR), K-Nearest Neighbors (KNN), Decision Tree (DT), Random Forest (RF), and Support Vector Machine (SVM) with linear kernel. LR models probability using a sigmoid function, suitable for linear decision boundaries. KNN with $k = 5$ classifies based on majority of nearest neighbors, but is sensitive to feature scale and noise. DT uses the CART algorithm, but tends to overfit due to lack of tree aggregation. In contrast, RF ($n_estimators = 100$) is a bagging ensemble that combines many trees with random feature subsets, reducing variance and being more robust to overfitting. Linear SVM seeks a maximum-margin separating hyperplane and is effective on linearly separable data. All models were evaluated using StratifiedKFold with $k = 5$ to maintain class balance, with metrics AUC, F1, accuracy, precision, and recall. To test performance stability, data was split into 80:20 and 70:30 ratios.

D. Hyperparameter Tuning and Threshold Analysis

RF was tuned using GridSearchCV with 5-fold cross-validation (cv=5) and AUC as the scoring metric (scoring='roc_auc') over the grid: max_depth [10,20,30,None], n_estimators [100,200,300], min_samples_leaf [1,2,4], min_samples_split [2,5,10], max_features ['sqrt','log2'] (216 combinations, 1080 fits per split). Subsequently, a threshold sweep from 0.30 to 0.70 (step 0.05) was performed on the test probabilities of the best model to evaluate performance across different decision boundaries, which is critical for clinical applications where the cost of false negatives is high. For each threshold, recall, precision, F1, and false negatives (FN) were computed, allowing us to select a threshold that balances sensitivity and specificity according to clinical priorities.

III. Results and Discussion

A. Baseline Comparison

Tables I and II present the performance of the five models. LR and SVM achieved the highest AUC (0.7960 and 0.7958 on the 80:20 split), but RF had the highest recall (0.6900) on both splits. Recall was chosen as the primary metric because in medical screening, False Negatives (undetected CVD patients) are far more dangerous than False Positives (unnecessary referrals). Theoretically, LR and SVM are optimal for linear data, but medical datasets are nonlinear; DT tends to overfit, KNN is sensitive to scale and noise, while RF as an ensemble of many trees with random feature subsets reduces variance and generalizes better to heterogeneous data (as evidenced by [3]). The consistency of rankings across splits (ΔAUC <0.002) confirms that the results are not partition artifacts, as shown in Figure 2.

TABLE I
BASELINE PERFORMANCE ON 80:20 SPLIT

Model	AUC	F1	Accuracy	Precision	Recall
LR	0.7960	0.7115	0.7315	0.7599	0.6689
KNN	0.7438	0.6861	0.6952	0.6995	0.6733
DT	0.6339	0.6274	0.6343	0.6329	0.6220
RF	0.7727	0.7033	0.7118	0.7170	**0.6900**
SVM	0.7958	0.7113	0.7307	0.7577	0.6702

TABLE II
BASELINE PERFORMANCE ON 70:30 SPLIT

Model	AUC	F1	Accuracy	Precision	Recall
LR	0.7930	0.7098	0.7289	0.7549	0.6697
KNN	0.7452	0.6901	0.6984	0.7020	0.6785
DT	0.6362	0.6300	0.6360	0.6339	0.6261
RF	0.7712	0.7031	0.7116	0.7167	**0.6900**
SVM	0.7928	0.7089	0.7277	0.7526	0.6700

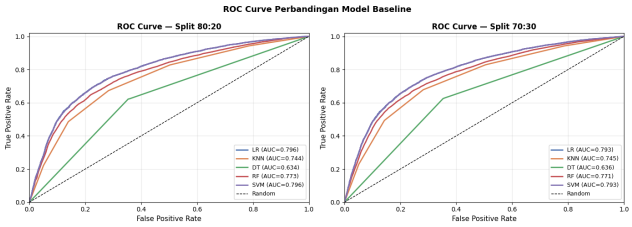

Fig. 2. ROC curves for the five models (left: 80:20, right: 70:30). LR and SVM have the highest AUC, but RF was chosen for its superior recall.

B. Hyperparameter Tuning

The optimal RF configuration (identical on both splits) was: max_depth=10, n_estimators=300, min_samples_leaf=4, min_samples_split=10, max_features='sqrt'. Best CV AUC = 0.7992 (80:20 split). The value max_depth=10 prevents overfitting by limiting tree depth; n_estimators=300 reduces ensemble variance; min_samples_leaf=4 improves leaf probability calibration.

Table III shows the improvement in AUC from 0.7727 to 0.8057 (80:20 split) and accuracy to 74.06%. Feature importance (Figure 3) shows that systolic pressure (ap_hi) is the strongest predictor (33.5%), followed by ap_lo (18.7%) and pulse pressure (15.8%). Together, these three blood pressure features contribute 68% of the model's decisions.

TABLE III
RF PERFORMANCE BEFORE AND AFTER TUNING

Split	Metric	Before	After	Change
80:20	AUC	0.7727	0.8057	+0.0330
	Accuracy	71.18%	74.06%	+2.88%
70:30	AUC	0.7712	0.8027	+0.0315
	Accuracy	71.16%	73.82%	+2.66%

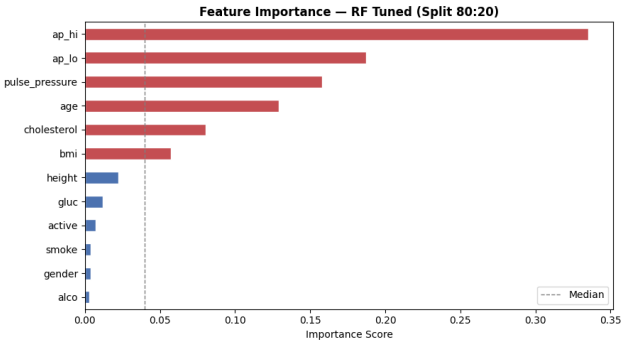

Fig. 3. Feature importance of tuned RF. Systolic pressure (ap_hi) dominates, followed by diastolic (ap_lo) and pulse pressure.

C. Threshold Analysis and Clinical Recommendation

Table IV shows that the threshold of 0.40 is more optimal than the default 0.50. At threshold 0.40, FN decreases from 2,183 to 1,517 (a reduction of 30.5%) with maximum F1 of 0.7408, and recall increases from 68 to 78 out of 100 referred patients. Figure 4 shows the rescue of 666 patients at the cost of 800 additional false positives, and this trade-off is acceptable because missed diagnosis is more dangerous.

TABLE IV
THRESHOLD SWEEP (TUNED RF, 80:20 SPLIT)

Threshold	Recall	Precision	F1	FN
0.30	0.8714	0.6387	0.7371	875
0.35	0.8308	0.6658	0.7392	1,151
0.40	**0.7770**	**0.7079**	**0.7408**	**1,517**
0.50	0.6792	0.7697	0.7216	2,183
0.60	0.5823	0.8152	0.6794	2,842

The model was tested on 10 random samples from the test data (5 CVD patients, 5 healthy). Table V shows that with threshold 0.40, the model correctly predicted 9 out of 10 cases. One error occurred for patient #3 (age 50, normal blood pressure) who actually had CVD but was predicted healthy (probability 19.5%). This result confirms that the model can operate as a clinical decision support tool.

TABLE V
INFERENCE SIMULATION ON 10 PATIENT SAMPLES

#	Age	SBP	DBP	BMI	Risk%	Pred(0.4)	Actual
1	55	150	90	32.5	85.6%	CVD	CVD
2	46	150	80	26.4	86.9%	CVD	CVD
3	50	100	70	28.2	19.5%	Healthy	CVD
4	42	130	80	31.2	56.0%	CVD	CVD
5	63	150	100	31.2	82.9%	CVD	CVD
6	40	120	80	23.4	17.2%	Healthy	Healthy
7	58	120	80	25.4	39.6%	Healthy	Healthy
8	46	120	70	23.6	24.8%	Healthy	Healthy
9	56	110	70	29.7	30.0%	Healthy	Healthy
10	39	120	80	25.6	17.8%	Healthy	Healthy

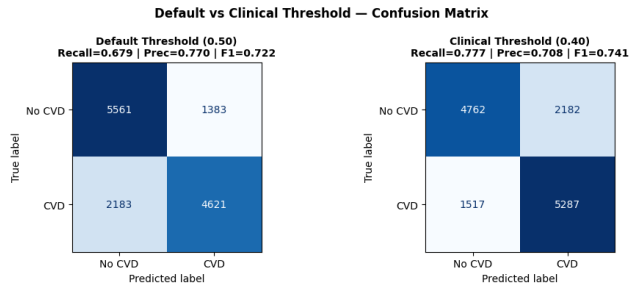

Fig. 4. Confusion matrix at default threshold 0.50 (left) and clinical threshold 0.40 (right). Lowering the threshold saves 666 CVD patients.

The screening system at Dr. Siti Rahmawati's clinic should use a threshold of 0.40, where patients with predicted CVD probability ≥ 0.40 are referred to a cardiologist, and the rest are monitored every 3 months. The 30.5% reduction in false negatives is equivalent to saving about 4.8% of patients who would otherwise go undetected.

IV. CONCLUSION

This study built a complete ML pipeline for CVD classification. Feature selection (16→12) preserved performance (AUC 0.7904). Among the five baselines, RF was chosen for its highest recall (0.6900) and robustness to overfitting due to bagging and random subspace. Tuning with GridSearchCV increased AUC to 0.8057. Threshold analysis recommends 0.40 to reduce false negatives by 30.5%. All pipeline stages were carried out according to the task framework. Limitations: the dataset does not include family history or actual ECG data. Future work could add SMOTE and SHAP.

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